

Practice Questions for Klee-Minty Problem

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1 Questions

1. The Klee-Minty problem demonstrates that the simplex method can exhibit exponential time complexity in the worst case. What is the fundamental characteristic of the Klee-Minty problem that leads to this behavior?
 - a. The presence of many degenerate vertices in the feasible region.
 - b. The use of a non-standard simplex method variant.
 - c. The highly skewed geometry of its feasible region, causing the simplex method to traverse many vertices before reaching the optimal solution.
 - d. The objective function's lack of linearity.
 - e. The problem's inherent non-convexity.
2. In the context of the Klee-Minty problem, what is the significance of the "highly skewed" nature of its feasible region?
 - a. It leads to numerical instability in the simplex method.
 - b. It causes the simplex method to cycle indefinitely.
 - c. It results in a large number of iterations for the simplex method to reach the optimal solution, demonstrating its worst-case exponential time complexity.
 - d. It makes the problem unsuitable for solution by any linear programming algorithm.
 - e. It renders the problem non-linear, making it unsolvable by the simplex method.
3. How does the Klee-Minty problem challenge the claim that the simplex method is efficient for solving linear programming problems?
 - a. It proves that the simplex method is always inefficient.
 - b. It demonstrates that the simplex method can require an exponential number of iterations in the worst case, contradicting the average-case polynomial behavior observed in practice.
 - c. It shows that the simplex method is only efficient for small-scale problems.
 - d. It highlights the limitations of using slack variables in the simplex method.
 - e. It proves that the simplex method is fundamentally flawed and should not be used.
4. What is the primary implication of the Klee-Minty problem's exponential worst-case complexity for the simplex method?

- a. The simplex method should be abandoned in favor of interior-point methods.
 - b. It proves that the simplex method is practically useless for large-scale problems.
 - c. It highlights a theoretical limitation of the simplex method, showing that its worst-case performance is not polynomial, even though its average-case performance is often observed to be polynomial in practice.
 - d. It indicates that the simplex method is inherently unstable and prone to numerical errors.
 - e. It demonstrates that the simplex method is not guaranteed to converge to an optimal solution.
5. The Klee-Minty problem is a notorious example demonstrating the simplex method's worst-case exponential time complexity. Explain the key features of the Klee-Minty problem that lead to this exponential behavior, and contrast this with the simplex method's typically observed polynomial average-case performance.
 - a. The Klee-Minty problem uses a highly skewed feasible region, forcing the simplex method to visit almost all vertices before reaching the optimal solution. This contrasts with the average-case polynomial behavior observed in practice, where the simplex method tends to find the optimal solution much faster.
 - b. The Klee-Minty problem utilizes a large number of variables and constraints, overwhelming the simplex method's computational capacity. This is in contrast to the average-case polynomial behavior, which is only observed for problems with a relatively small number of variables and constraints.
 - c. The Klee-Minty problem's objective function is non-linear, causing the simplex method to struggle. This differs from the average-case polynomial behavior, which is only observed for linear objective functions.
 - d. The Klee-Minty problem introduces degeneracy, causing cycling in the simplex method. This contrasts with the average-case polynomial behavior, which assumes non-degenerate problems.
 - e. The Klee-Minty problem's constraints are ill-conditioned, leading to numerical instability in the simplex method. This is different from the average-case polynomial behavior, which assumes well-conditioned problems.
6. The Klee-Minty problem is a specific linear program designed to demonstrate what characteristic of the simplex method?
 - a. Its average-case polynomial time complexity.
 - b. Its ability to handle degenerate cases efficiently.
 - c. Its worst-case exponential time complexity.
 - d. Its effectiveness in solving large-scale problems.
 - e. Its insensitivity to the initial starting point.
7. What is a key feature of the feasible region in the Klee-Minty problem that contributes to the simplex method's exponential runtime?
 - a. It is a convex polygon.
 - b. It is a highly skewed polytope.
 - c. It has a small number of vertices.

- d. It is unbounded.
 - e. It is centrally symmetric.
8. In the Klee-Minty problem, how does the simplex method's path to the optimal solution relate to the geometry of the feasible region?
- a. It follows a shortest path between vertices.
 - b. It visits every vertex of the feasible region.
 - c. It follows a path that is independent of the feasible region's shape.
 - d. It visits a subset of vertices, determined by the objective function.
 - e. It traces the boundary of the feasible region.
9. What is the primary implication of the Klee-Minty problem's demonstration of exponential worst-case complexity for the simplex method?
- a. The simplex method should never be used.
 - b. The simplex method is practically unusable for large problems.
 - c. There exist linear programming problems for which the simplex method can be extremely inefficient.
 - d. All linear programming problems require exponential time to solve.
 - e. Polynomial-time algorithms for linear programming do not exist.
10. In the Klee-Minty problem (Slide 2), what is the coefficient of x_1 in the third constraint?
- a. 1
 - b. 2
 - c. 1023
 - d. 2047
 - e. 0

2 Answers

1. The Klee-Minty problem demonstrates that the simplex method can exhibit exponential time complexity in the worst case. What is the fundamental characteristic of the Klee-Minty problem that leads to this behavior?
 - a. While degeneracy can slow down the simplex method, it doesn't inherently cause exponential time complexity. The Klee-Minty problem's exponential behavior is due to the geometry, not degeneracy.
 - b. The Klee-Minty problem uses a standard simplex method. The exponential behavior is a property of the problem itself, not a flaw in a specific variant of the algorithm.
 - c. The Klee-Minty problem is specifically constructed to have a feasible region that is a highly distorted polytope. The simplex method, following its rules, will visit a large number of vertices, resulting in exponential time complexity. The skewed geometry forces the algorithm to take a long path to the optimal solution.
 - d. The objective function in the Klee-Minty problem is linear. The exponential complexity arises from the structure of the constraints and the resulting geometry of the feasible region.
 - e. The feasible region of a linear program is always convex. The Klee-Minty problem is a linear program, and its feasible region is convex. The exponential complexity is a result of the specific shape of this convex region.
2. In the context of the Klee-Minty problem, what is the significance of the "highly skewed" nature of its feasible region?
 - a. While numerical instability can be a concern in linear programming, it's not the primary reason for the Klee-Minty problem's exponential complexity. The problem's structure is the main cause.
 - b. Cycling is a potential issue in the simplex method, but the Klee-Minty problem doesn't cause cycling. It demonstrates exponential runtime without cycling.
 - c. The highly skewed geometry of the Klee-Minty problem's feasible region is the key factor that causes the simplex method to take an exponentially long time to find the optimal solution. The algorithm is forced to visit a large number of vertices before reaching the optimum.
 - d. The Klee-Minty problem is solvable by the simplex method, but it takes an exponentially long time in the worst case. Other algorithms might solve it faster.
 - e. The Klee-Minty problem is a linear programming problem; its objective function and constraints are linear. The difficulty stems from the geometry of the feasible region, not the linearity of the problem.
3. How does the Klee-Minty problem challenge the claim that the simplex method is efficient for solving linear programming problems?
 - a. The Klee-Minty problem doesn't prove that the simplex method is always inefficient. It shows that it can be inefficient in specific, carefully constructed cases.
 - b. The Klee-Minty problem provides a specific example where the simplex method, while often efficient in practice, can exhibit exponential time complexity in the worst case. This contrasts with its observed average-case polynomial behavior, highlighting a theoretical limitation.
 - c. The simplex method is widely used and efficient for many real-world problems, even large-scale ones. The Klee-Minty problem is a specific counterexample, not a general limitation.

- d. The Klee-Minty problem's exponential complexity is not directly related to the use of slack variables. The issue lies in the geometry of the feasible region.
 - e. The simplex method is a widely used and valuable algorithm. The Klee-Minty problem highlights a theoretical worst-case scenario, not a fundamental flaw.
4. What is the primary implication of the Klee-Minty problem's exponential worst-case complexity for the simplex method?
- a. While interior-point methods offer polynomial time complexity, the Klee-Minty problem doesn't necessitate abandoning the simplex method. It's still widely used and effective in many applications.
 - b. The simplex method is successfully used for many large-scale problems. The Klee-Minty problem highlights a theoretical worst-case scenario, not a practical limitation for all large problems.
 - c. The Klee-Minty problem's exponential worst-case complexity is a theoretical result. It doesn't mean the simplex method is always slow or unreliable, but it does show that its worst-case runtime is not bounded by a polynomial function of the input size, despite its generally good performance in practice.
 - d. Numerical instability can occur in linear programming, but it's not the direct cause of the Klee-Minty problem's exponential complexity. The problem's structure is the key factor.
 - e. The simplex method, when implemented correctly, is guaranteed to converge to an optimal solution for linear programs. The Klee-Minty problem shows it can take a very long time to do so in certain cases.
5. The Klee-Minty problem is a notorious example demonstrating the simplex method's worst-case exponential time complexity. Explain the key features of the Klee-Minty problem that lead to this exponential behavior, and contrast this with the simplex method's typically observed polynomial average-case performance.
- a. The Klee-Minty problem's construction deliberately creates a highly skewed feasible region, forcing the simplex method to traverse a large number of vertices, resulting in exponential time complexity in the worst case. This contrasts sharply with the empirically observed average-case polynomial behavior, where the simplex method typically performs much more efficiently.
 - b. While the Klee-Minty problem can be scaled to have many variables and constraints, the exponential behavior is not solely due to the problem size but rather the specific geometry of the feasible region. The average-case polynomial behavior is observed across a wide range of problem sizes.
 - c. The Klee-Minty problem uses a linear objective function. The exponential behavior arises from the structure of the constraints, not the nature of the objective function. The average-case polynomial behavior also applies to problems with linear objective functions.
 - d. While degeneracy can lead to cycling in the simplex method, the Klee-Minty problem does not rely on degeneracy to achieve its exponential behavior. The average-case polynomial behavior is often observed even in the presence of some degeneracy.
 - e. The Klee-Minty problem does not inherently involve ill-conditioned constraints. The exponential behavior is a consequence of the specific geometric properties of the feasible region, not numerical instability. The average-case polynomial behavior is observed even for problems with well-conditioned constraints.
6. The Klee-Minty problem is a specific linear program designed to demonstrate what characteristic of the simplex method?

- a. While empirical studies suggest the simplex method performs well on average, the Klee-Minty problem is a counterexample to average-case polynomial time.
 - b. The Klee-Minty problem does not directly address the efficiency of handling degenerate cases.
 - c. The Klee-Minty problem is specifically constructed to force the simplex method to take an exponential number of iterations, thus showcasing its worst-case exponential time complexity.
 - d. The Klee-Minty problem highlights a limitation, not an advantage, in solving large-scale problems.
 - e. The simplex method's path to the optimal solution is highly dependent on the initial starting point, and the Klee-Minty problem exploits this sensitivity.
7. What is a key feature of the feasible region in the Klee-Minty problem that contributes to the simplex method's exponential runtime?
- a. The feasible region is indeed a polytope (a generalization of a polygon to higher dimensions), but its convexity is not the primary reason for the exponential runtime.
 - b. The Klee-Minty problem's feasible region is deliberately constructed to be highly skewed, causing the simplex method to traverse many vertices before reaching the optimal solution.
 - c. The Klee-Minty problem's feasible region has a large number of vertices, contributing to the exponential number of iterations.
 - d. The Klee-Minty problem's feasible region is bounded.
 - e. The feasible region is not centrally symmetric; its asymmetry is crucial to the problem's design.
8. In the Klee-Minty problem, how does the simplex method's path to the optimal solution relate to the geometry of the feasible region?
- a. The simplex method does not necessarily follow the shortest path; it can take a much longer route due to the problem's construction.
 - b. The simplex method does not visit every vertex in the Klee-Minty problem, but it does visit a number of vertices that grows exponentially with the problem size.
 - c. The path taken by the simplex method is highly dependent on the shape and structure of the feasible region, which is precisely what the Klee-Minty problem exploits.
 - d. The simplex method, when applied to the Klee-Minty problem, visits a specific subset of vertices, determined by the objective function and the highly skewed nature of the feasible region. This subset grows exponentially with the problem size.
 - e. The simplex method does not necessarily trace the boundary of the feasible region; it moves from vertex to vertex in the interior of the region.
9. What is the primary implication of the Klee-Minty problem's demonstration of exponential worst-case complexity for the simplex method?
- a. The simplex method remains a widely used and effective algorithm for linear programming despite its worst-case exponential complexity.
 - b. While the Klee-Minty problem demonstrates potential inefficiency for large problems, the simplex method's average-case performance remains polynomial, making it practically useful.

- c. The Klee-Minty problem shows that while the simplex method is efficient in practice for most problems, there are specific instances where it can exhibit exponential runtime, highlighting a theoretical worst-case scenario.
 - d. The existence of polynomial-time algorithms for linear programming (like the ellipsoid method and interior-point methods) refutes this statement.
 - e. The existence of polynomial-time algorithms for linear programming contradicts this claim.
10. In the Klee-Minty problem (Slide 2), what is the coefficient of x_1 in the third constraint?
- a. The coefficient of x_1 in the third constraint of the Klee-Minty problem is 1. The constraints are designed to have a specific structure that leads to exponential behavior in the simplex method.
 - b. Incorrect. This is not the coefficient of x_1 in the third constraint.
 - c. Incorrect. This is a large number, but not the coefficient of x_1 in the third constraint of the Klee-Minty problem presented.
 - d. Incorrect. This is a large number, but not the coefficient of x_1 in the third constraint of the Klee-Minty problem presented.
 - e. Incorrect. The coefficient of x_1 is not 0; it's a key part of the problem's structure.